SPSS Classes

Basics, Descriptive Statistics, T-Tests and One-Way ANOVA

Part 1 1: SPSS Basics & Descriptive Statistics
Part 2: T-Tests and ANOVA

Simple examples **SPSS menus** **Syntax** **Interpretation**

Md Salek Miah
Statistics, SUST

Class Plan — Serial Order

Start with SPSS basics, then move to t-tests and ANOVA.

CLASS 1

SPSS Basics & Descriptive Statistics

- SPSS interface and workflow
- Variable types, coding, labeling
- Data entry and import
- Frequency tables
- Mean, median, mode
- SD, variance, range

CLASS 2

T-Tests and ANOVA

- One-sample t-test
- Independent-sample t-test
- Paired-sample t-test
- One-way ANOVA
- Tukey post-hoc tests
- Interpretation of output

SPSS Class 1

SPSS Basics and Descriptive Statistics

Interface • Variable View • Coding • Data entry • Frequencies • Mean/Median/Mode • SD/Variance/Range

ID	sex	age	score
1	1	20	72
2	2	22	68
3	1	21	80
4	2	23	75
5	1	19	70

Data View

Learning Objectives

By the end of Class 1, students can prepare data and summarize variables in SPSS.

After this class, you can:

- Identify variable type and measurement level
- Code and label values correctly
- Enter data and import Excel/CSV files
- Produce frequency tables
- Interpret mean, median, mode and spread

SPSS workflow

1. **Open data**
2. **Check Variable View**
3. **Run Analyze menu**
4. **Read Output**

Main rule: choose the correct statistic from the variable type first, not from the p-value.

SPSS Interface and Workflow

Know where to enter data, define variables, write syntax, and read results.

Data View

Rows = cases
Columns = variables
Use for data entry

Variable View

Name, type, labels
values, missing, measure
Use before analysis

Output Viewer

Tables and graphs appear here
Copy results to Word

Syntax Editor

Saves commands
Makes analysis reproducible

Variable Types, Coding and Labeling

Before analysis, every variable needs a clear name, code, label and measure level.

SPSS Measure	Meaning	Example	Common summary
Nominal	Categories, no order	Sex, residence, division	Frequencies, percentages
Ordinal	Categories with order	Education level, wealth quintile	Frequencies, median
Scale	Numeric continuous/discrete	Age, BMI, score, income	Mean, SD, min–max
String vs Numeric	Text or number storage	Name = string; age = numeric	Usually analyze numeric codes

Tip: Use numeric codes with value labels. Example: residence 1 = Urban, 2 = Rural.

Coding and Labeling — Example

Good coding makes SPSS output easy to read and reduces data-entry mistakes.

Variable	Type	Values / labels	Missing	Measure
id	Numeric	Unique ID	None	Nominal
sex	Numeric	1 = Male; 2 = Female	9 = Missing	Nominal
education	Numeric	1 = Primary; 2 = Secondary; 3 = Higher	9 = Missing	Ordinal
score	Numeric	0–100 test score	999 = Missing	Scale

Variable name

Short, no spaces: study_hours, final_score

Value label

Makes output show categories instead of only numbers.

Data Entry and Import

SPSS can use manually entered data or imported Excel/CSV datasets.

Manual data entry

- Each row = one participant/case
- Each column = one variable
- Enter only valid codes
- Check missing codes before analysis

Import Excel or CSV

- File → Import Data → Excel/CSV
- Tick “Read variable names” if first row has names
- Check Variable View after import
- Save as .sav for future work

Frequency Tables

Use frequencies for categorical variables and to check data quality.

When to use

Nominal and ordinal variables

Counts and percentages

Check impossible values

Describe sample characteristics

SPSS menu

Analyze → Descriptive Statistics → Frequencies

Click OK

Syntax

```
FREQUENCIES VARIABLES=sex residence education
  /ORDER=ANALYSIS.
```

Output: Frequency, Percent, Valid Percent, Cumulative Percent.

Mean, Median and Mode

These measures describe the centre of a distribution.

Mean

Average value

Best for roughly symmetric scale data

Median

Middle value

Useful when data are skewed or have outliers

Mode

Most frequent value

Useful for categories and repeated values

SPSS path: Analyze → Descriptive Statistics → Frequencies → Statistics → Mean, Median, Mode

Standard Deviation, Variance and Range

These measures describe how spread out the values are.

Statistic	Simple meaning	How to interpret
Standard deviation (SD)	Typical distance from the mean	Large SD = values are more spread out
Variance	SD squared	Useful mathematically; less intuitive for reporting
Range	Maximum − minimum	Shows total spread but is sensitive to outliers

Example

Mean score = 70, SD = 8 means most scores are around 70 with moderate variation.

SPSS path

Analyze → Descriptive Statistics → Descriptives → Options

Descriptive Statistics — Command + Syntax

Run descriptives from the SPSS menu or save the commands as syntax.

SPSS Menu Command

1. Analyze → Descriptive Statistics → Descriptives
2. Move scale variables to Variable(s)
3. Click Options
4. Choose mean, SD, variance, range, min, max
5. Click Continue → OK

SPSS Syntax

```
DESCRIPTIVES VARIABLES=age score
  /STATISTICS=MEAN STDDEV VARIANCE RANGE MIN
MAX.

FREQUENCIES VARIABLES=sex residence
  /ORDER=ANALYSIS.
```

Syntax helps you repeat the same analysis correctly.

How to Interpret Descriptive Output

Look at valid N, missing values, centre, spread and possible errors.

Variable	N	Mean	Median	SD	Min	Max
score	30	72.4	73.0	8.6	55	90

Interpretation sentence

“The mean test score was 72.4 (SD = 8.6), with scores ranging from 55 to 90.”

Do not report only one number. Give centre and spread together.

Quick practice

Which variables need frequency tables?

Which variables need mean and SD?

Are there impossible values?

Are missing codes correctly defined?

Class 1 Summary and Link to Class 2

Descriptive statistics prepare the data before inferential tests.

Class 1 summary

- Define variables correctly
- Use labels and missing values
- Use frequencies for categories
- Use mean/median/mode for centre
- Use SD/variance/range for spread

Next: Class 2

- One-sample t-test
- Independent-sample t-test
- Paired-sample t-test
- One-way ANOVA with Tukey
- Output interpretation

t

F

SPSS Class 2

T-Tests and One-Way ANOVA

One-sample • Independent-sample • Paired-sample • Tukey post-hoc • Output interpretation

Simple Examples | Commands | Syntax | Results

Md Salek Miah
Statistics, SUST

Simple examples • SPSS commands • Syntax • Result interpretation

Learning Objectives

By the end of the class, students should be able to choose, run and interpret the correct test.

After this class, you can:

- Select the correct t-test or one-way ANOVA
- Run the test using SPSS menu commands
- Run the same test using SPSS syntax
- Read p-values, confidence intervals and key output
- Use Tukey post-hoc after a significant ANOVA

Which test should I use?

1 sample mean	vs fixed value	**One-sample t-test**
2 separate groups	compare means	**Independent t-test**
Same people twice	before / after	**Paired t-test**
3+ independent groups	compare means	**One-way ANOVA**

Before Running the Tests

Check the design and the basic assumptions first.

1
Outcome variable

Continuous / scale

e.g., score, weight, blood pressure

2
Independence

Cases should be independent

Except paired t-test, where pairs are related

3
Distribution

No serious outliers; roughly normal

Especially important with small samples

4
Equal variances

For independent t-test / ANOVA

Check Levene's test

Tip: A statistically significant result ($p < .05$) does not automatically mean the difference is practically important.

One-Sample t-Test

Compare one sample mean with one fixed or expected value.

Simple idea

ONE group
vs
ONE benchmark

Bangladeshi example: A coaching centre says students study 4 hours/day. Is the average study time of 30 SUST students different from 4 hours?

Study design

Outcome: study_hours (scale)
Benchmark/Test value: 4 hours
Only one sample of students.

Hypotheses

H_0: $\mu = 4$ hours

H_1: $\mu \neq 4$ hours

Decision

If $p < .05$ → evidence of a difference.

Why this test?

We are not comparing two groups. We have only one group, and we want to compare its average with a fixed number (4 hours).

সহজভাবে: ১টি group-এর average কি নির্দিষ্ট মান থেকে আলাদা?

One-Sample t-Test — Command + Syntax

Run the same test either from the menu or with syntax.

SPSS Menu Command

1. **Open the test**
   Analyze → Compare Means → One-Sample T Test
2. **Move variable**
   Put study_hours into Test Variable(s).
3. **Enter benchmark**
   Type 4 into Test Value.
4. **Run**
   Click OK.

SPSS Syntax

```
T-TEST
  /TESTVAL = 4
  /MISSING = ANALYSIS
  /VARIABLES = study_hours
  /CRITERIA = CI(.95).
```

Meaning: compare the mean of study_hours with 4 hours.

One-Sample t-Test — Simple Result Example

Translate the SPSS numbers into an everyday conclusion.

Real-life example

30 students are surveyed. Their average study time is 4.6 hours/day. We test whether this differs from 4 hours/day.

SPSS gives

Mean = 4.60
Test value = 4.00
$t(29) = 2.35$
$p = .026$

Explain it simply

Because $p = .026$ is below .05, the difference is unlikely to be due to random sampling alone.

Significant difference

How to report

"Students studied significantly more than 4 hours per day on average, $t(29) = 2.35$, $p = .026$."

Do not write only "$p < .05$".

One-Sample t-Test — Output Interpretation

Read the descriptive statistics first, then the p-value and confidence interval.

Example output

Mean	Test Value	t	df	Sig. (2-tailed)	Mean Difference
4.60	4.00	2.35	29	.026	0.60

How to interpret

p = .026, which is less than .05 → statistically significant.
The sample mean (4.60) is significantly different from 4.00.
Mean difference = 0.60 hour.

Report example: "Mean daily study time was significantly different from 4 hours, t(29) = 2.35, p = .026."

Independent-Samples t-Test

Compare the means of two separate, unrelated groups.

Simple idea

TWO different groups
Group A vs Group B

Bangladeshi example: Do SUST students living in a hall and students living outside the hall have different average daily study hours?

Study design

Outcome: study_hours (scale)
Group: residence
1 = Hall, 2 = Outside
Different students in each group.

Hypotheses

H_0: μHall = μOutside

H_1: μHall ≠ μOutside

Decision

If $p < .05$ → evidence of a difference.

Why this test?

Each student belongs to only one group. The two groups are separate, so the observations are independent.

সহজভাবে: ২টি আলাদা group-এর average কি আলাদা?

Independent t-Test — Command + Syntax

Define the outcome and the two independent groups.

SPSS Menu Command

1. **Open the test**
   Analyze → Compare Means → Independent-Samples T Test
2. **Outcome**
   Move study_hours to Test Variable(s).
3. **Grouping variable**
   Move residence to Grouping Variable.
4. **Define groups**
   Group 1 = 1; Group 2 = 2.
5. **Run**
   Click Continue → OK.

SPSS Syntax

```
T-TEST GROUPS = residence(1 2)
  /MISSING = ANALYSIS
  /VARIABLES = study_hours
  /CRITERIA = CI(.95).
```

Coding example: 1 = Hall student, 2 = Outside-hall student.

Independent t-Test — Simple Result Example

First check Levene's test, then interpret the t-test p-value.

Real-life example

Hall students: mean study time = 4.8 hours. Outside-hall students: mean = 4.1 hours. Are the two group averages different?

SPSS gives

Levene p = .420
t(58) = 2.12
p = .038
Mean difference = 0.70 hour
0,70(0.68-0.78)

Explain it simply

Levene p > .05, so use the "Equal variances assumed" row. Then p = .038 < .05, so the two group means differ.

Significant group difference

How to report

"Hall and outside-hall students differed significantly in daily study time, t(58) = 2.12, p = .038."

Do not write only "p < .05".

Independent t-Test — Output Interpretation

Use Levene's test first, then choose the correct t-test row.

Step 1: Levene's test

If Levene p > .05 → use "Equal variances assumed"
If Levene p < .05 → use "Equal variances not assumed"

Example output

Levene p	t	df	Sig. (2-tailed)	Mean Difference
.420	2.12	58	.038	0.70

Levene p = .420 > .05 → use the "Equal variances assumed" row.
t-test p = .038 < .05 → the two group means are significantly different.
Mean difference = 0.70 hour.

Report example: "Hall students scored significantly higher than outside-hall students, t(58) = 2.12, p = .038."

Paired-Samples t-Test

Compare two measurements from the same people or matched pairs.

Simple idea

SAME people
Before vs After

Bangladeshi example: The same 25 students take an SPSS quiz before and after a two-week training course. Did their scores improve?

Study design

Outcome 1: pre_score
Outcome 2: post_score
The SAME 25 students provide both scores.

Hypotheses

H_0: Mean difference = 0

H_1: Mean difference ≠ 0

Decision

If $p < .05$ → evidence of a difference.

Why this test?

The two scores are linked person-by-person. We care about how each student changed, not about two separate groups.

সহজভাবে: একই মানুষের before–after average কি বদলেছে?

Paired t-Test — Command + Syntax

Create a pair using the two measurements from the same participants.

SPSS Menu Command

1. **Open the test**
   Analyze → Compare Means → Paired-Samples T Test

2. **Select variables**
   Select pre_score and post_score.

3. **Create Pair 1**
   Move both variables into Pair 1.

4. **Run**
   Click OK.

SPSS Syntax

```
T-TEST PAIRS = pre_score WITH
post_score (PAIRED)
  /CRITERIA = CI(.95)
  /MISSING = ANALYSIS.
```

SPSS calculates the difference in the order shown: pre_score − post_score.

Paired t-Test — Simple Result Example

Think about change within the same students.

Real-life example

The same 25 students score 65.2 before training and 72.8 after training. We test whether the average change is statistically significant.

SPSS gives

Pre mean = 65.2
Post mean = 72.8
$t(24) = -4.30$
$p < .001$

Explain it simply

The post-training average is higher. Because $p < .001$, the improvement is statistically significant.

Significant improvement

How to report

"Quiz scores increased significantly after the SPSS training, $t(24) = -4.30$, $p < .001$."

Do not write only "$p < .05$".

Paired t-Test — Output Interpretation

Focus on the mean difference, t, df and Sig. (2-tailed).

Pre Mean	Post Mean	Mean Diff. (Pre–Post)	t	df	Sig.
65.20	72.80	−7.60	−4.30	24	< .001

Interpretation

- p < .001 → the pre and post means are significantly different.
- Post mean (72.80) is higher than pre mean (65.20).
- The negative mean difference occurs because SPSS calculated Pre − Post.

Report example: “Scores increased significantly after training, t(24) = −4.30, p < .001.”

One-Way ANOVA

Compare the means of three or more independent groups.

Simple idea

THREE or more groups
A vs B vs C

Bangladeshi example: An institute teaches SPSS in three ways—Lecture, Video, and Group Practice. Do the three groups have different average final test scores?

Study design

Outcome: final_score (scale)
Factor: method
1 = Lecture, 2 = Video, 3 = Group Practice
Different students in each group.

Hypotheses

H_0: $\mu1 = \mu2 = \mu3$

H_1: At least one mean differs

Decision

If $p < .05$ → evidence of a difference.

Why this test?

ANOVA answers one overall question for 3+ groups. Running many separate t-tests would increase the chance of a false positive result.

সহজভাবে: ৩ বা তার বেশি group-এর average কি এক?

One-Way ANOVA — Command + Syntax

Run ANOVA, check homogeneity, and request Tukey post-hoc.

SPSS Menu Command

1. **Open ANOVA**
   Analyze → Compare Means → One-Way ANOVA
2. **Set variables**
   final_score → Dependent List; method → Factor
3. **Options**
   Choose Descriptives + Homogeneity of variance test.
4. **Post Hoc**
   Choose Tukey → Continue.
5. **Run**
   Click OK.

SPSS Syntax

```
ONEWAY final_score BY method
  /STATISTICS DESCRIPTIVES
HOMOGENEITY
  /MISSING ANALYSIS
  /POSTHOC = TUKEY ALPHA(0.05).
```

Tukey tells you WHICH groups differ after a significant ANOVA.

One-Way ANOVA — Simple Result Example

ANOVA first; Tukey second.

Real-life example

Mean final scores: Lecture = 65, Video = 72, Group Practice = 80. We ask whether these three averages are all the same.

SPSS gives

F(2, 57) = 8.41
p = .001
Tukey: Lecture vs Group p=.001
Video vs Group p=.040

Explain it simply

ANOVA p = .001 < .05, so at least one teaching method differs. Tukey then shows where the differences are.

Significant ANOVA

How to report

"Final scores differed significantly by teaching method, F(2, 57) = 8.41, p = .001. Tukey showed Group Practice scored higher than Lecture and Video."

Do not write only "p < .05".

One-Way ANOVA + Tukey — Output Interpretation

Interpret in three stages: descriptives → ANOVA → post-hoc.

1 Descriptives

Method	Mean
Lecture	65.0
Video	72.0
Group	80.0

Look at the pattern of group means.

2 ANOVA table

F	df	Sig.
8.41	2, 57	.001

p = .001 < .05 → at least one mean differs.

Now check Tukey.

3 Tukey post-hoc

Comparison	p
	.310
Lecture vs Group	.001
Video vs Group	.040

Significant: Lecture vs Group; Video vs Group

Not significant: Lecture vs Video

How to Report Results + Quick Summary

Use a short sentence that gives the test statistic, degrees of freedom and p-value.

One-sample t-test	Mean study time was significantly different from 4 hours, $t(29) = 2.35$, $p = .026$.
Independent t-test	Hall and outside-hall students differed in study time, $t(58) = 2.12$, $p = .038$.
Paired t-test	SPSS quiz scores increased after training, $t(24) = -4.30$, $p < .001$.
One-way ANOVA	Final scores differed by teaching method, $F(2, 57) = 8.41$, $p = .001$.

Remember: choose the test from the study design first — not from the p-value.

Quick Practice

Choose the correct test before opening SPSS.

1. Is the average daily study time of one student sample different from 4 hours?	**One-sample t-test**
2. Do hall and outside-hall students have different average study hours?	**Independent t-test**
3. The same students take a test before and after an SPSS training course.	**Paired t-test**
4. Final test scores are compared across Lecture, Video and Group Practice.	**One-way ANOVA**

Next class idea: correlation, chi-square or non-parametric alternatives.